ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Simultaneous Equations

Choose the method that removes one unknown fastest.

Simultaneous Equations · P1 · 03

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 ELIM: Eliminate A Letter

TRIGGER *two linear equations*

02 SUBS: Substitute Directly

TRIGGER *one equation already gives $x =$ or $y =$*

03 CURVE: Line Meets Curve

TRIGGER *linear and quadratic equation together*

04 CHECK: Check The Pair

TRIGGER *answer is an ordered pair*

BANK EVIDENCE Local pages: 27 problem, 48 answer, 9 notes. Website primary entries: 3.

01 ELIM: Eliminate A Letter

TRIGGER *two linear equations*

BECOMES Two equations become one equation in one unknown.

FIRST LINE TO WRITE

$$2x + 3y = 13$$
$$4x - y = 5$$

SIMPLEST STRATEGY

1. **E**qualise one coefficient.
2. **L**ine up the equations.
3. **I**dentify add or subtract.
4. **M**ake one letter disappear.

WORKED MODEL

$$2(2x + 3y = 13) \Rightarrow 4x + 6y = 26. \quad (4x + 6y) - (4x - y) = 21,\ y = 3.$$

HARD VARIANTS

1. If coefficients are awkward, multiply both equations until one letter has matching coefficients.
2. If equations contain fractions, clear denominators before eliminating.
3. After solving, substitute into both original equations, not the simplified copies only.

BOTTOM LINE

Elimination is safest when coefficients can be matched quickly.

Practice 01

SAME IDEA Use ELIM: Eliminate A Letter.

QUESTION

Solve $3x + 2y = 16,\ 2x - y = 3.$

YOUR SOLUTION

02 SUBS: Substitute Directly

TRIGGER *one equation already gives $x =$ or $y =$*

BECOMES One unknown is replaced by an expression.

FIRST LINE TO WRITE

$$y = 2x + 1$$

SIMPLEST STRATEGY

1. **S**elect the isolated equation.
2. **U**se it inside the other equation.
3. **B**uild one equation.
4. **S**ubstitute back.

WORKED MODEL

$$y = 2x + 1, \quad x + y = 10 \Rightarrow x + 2x + 1 = 10, \; x = 3, \; y = 7.$$

HARD VARIANTS

1. If nothing is isolated, isolate the variable with the smallest coefficient first.
2. When substitution creates a quadratic, keep both roots and find both coordinate pairs.
3. Reject any pair that fails one original equation.

BOTTOM LINE

Substitute when a variable is already isolated.

Practice 02

SAME IDEA Use SUBS: Substitute Directly.

QUESTION

Solve $y = 3x - 2, \ x + y = 14$.

YOUR SOLUTION

03 CURVE: Line Meets Curve

TRIGGER *linear and quadratic equation together*

BECOMES Substitute the line into the curve to get a quadratic.

FIRST LINE TO WRITE

$$y = mx + c \quad \text{goes into} \quad y = x^2 + ax + b$$

SIMPLEST STRATEGY

1. **C**hoose the line equation.
2. **U**se it in the curve.
3. **R**earrange to a quadratic.
4. **V**erify both coordinates.
5. **E**xpress both intersection points.

WORKED MODEL

$$y = x + 2,\ y = x^2 \Rightarrow x^2 = x + 2 \Rightarrow (x-2)(x+1) = 0.$$

HARD VARIANTS

1. Line-curve hard versions usually become a quadratic; keep both intersections.
2. For tangency, use the repeated-root condition or discriminant $= 0$.
3. Always turn each x-value back into a full coordinate.

BOTTOM LINE

Intersections are found by making the two expressions for y equal.

Practice 03

SAME IDEA Use CURVE: Line Meets Curve.

QUESTION

Find where $y = x + 6$ meets $y = x^2$.

YOUR SOLUTION

04 CHECK: Check The Pair

TRIGGER *answer is an ordered pair*

BECOMES The pair must satisfy both original equations.

FIRST LINE TO WRITE

$$(x, y) = (\cdot, \cdot)$$

SIMPLEST STRATEGY

1. **C**hoose the original equations.
2. **H**it both equations with the pair.
3. **E**valuate left and right sides.
4. **C**orrect if one equation fails.
5. **K**eep the ordered pair notation.

WORKED MODEL

$$(3, 7): \quad 3 + 7 = 10, \quad 7 = 2(3) + 1.$$

HARD VARIANTS

1. For parameter questions, check the pair before using it to find the parameter.
2. If a point is claimed to lie on two curves, test both equations separately.
3. State the ordered pair in the requested order.

BOTTOM LINE

A simultaneous answer is not finished until both equations agree.

Practice 04

SAME IDEA Use CHECK: Check The Pair.

QUESTION

> Check whether $(2, 5)$ solves $x + y = 7,\ 2x - y = -1$.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
two linear equations	$2x + 3y = 13$ $4x - y = 5$
one equation already gives $x =$ or $y =$	$y = 2x + 1$
linear and quadratic equation together	$y = mx + c$ goes into $y = x^2 + ax + b$
answer is an ordered pair	$(x, y) = (\cdot, \cdot)$